

Abstract Algebra II

Homework 5

Due: 6th May 2026

Submit only Questions 1 and 2.

1. Let K_n be the cyclotomic extension of $\mathbb{Q}$ obtained by adjoining a primitive n -th root of unity to $\mathbb{Q}$. Determine the structure of $\text{Aut}(K_n/\mathbb{Q})$.
2. Find all positive integers n such that the polynomial $x^{2^n} + x + 1$ is irreducible over $\mathbb{F}_2$.
3. Let K be a field of characteristic $p > 0$. For an element $a \in K \setminus K^p$, i.e., a is not a p -th power in K , find all integers n such that $x^{p^n} - a \in K[x]$ is irreducible.
4. Let K be a finite field. Show that every algebraic closure $\overline{K}$ of K is Galois over K .